

Leave Approval System

1. Introduction

The **Leave Approval System** is a ServiceNow-based application that automates the process of applying for, approving, and tracking employee leave requests. Instead of relying on manual paperwork or emails, employees can submit leave requests through a digital form, managers can approve or reject requests, and the system automatically updates the leave status and sends notifications. This automation improves efficiency, transparency, and accuracy in leave management.

1.1 Project Overview

Project Name : Leave Approval System

Project Version : 1.0

Project Type : Individual Project

Platform : ServiceNow Personal Developer Instance (PDI)

Testing Period : 26 June 2026 to 30 June 2026

Prepared By : SUBHA R

Project Description:

The Leave Approval System is designed to simplify leave management within an organization. Employees submit leave requests with details such as leave type and leave dates. The system automatically routes the request to the reporting manager using Flow Designer. Based on the manager's decision, the leave request is either approved or rejected, and the employee receives an email notification.

Key Components Built:

- Custom Leave Request Application developed using ServiceNow Studio.
- Leave Request Table with employee details, leave type, start date, end date, reason, and leave status fields.

- Flow Designer Workflow to automate the leave approval process by routing requests to the reporting manager.
- Manager Approval Process using the Ask for Approval action with Approved and Rejected branches.
- Email Notifications for:
 - Leave Request Submitted (Manager)
 - Leave Request Approved (Employee)
 - Leave Request Rejected (Employee)
- Business Rules for:
 - Calculating leave duration automatically.
 - Validating start and end dates.
 - Preventing duplicate leave requests.
- Leave Status Tracking with statuses such as Pending, Approved, and Rejected.
- ServiceNow Form Design with user-friendly fields for employees to submit leave requests.
- GitLab Repository created for source code version control and project management.
- Testing and Validation performed on the ServiceNow Personal Developer Instance (PDI) to verify the complete leave approval workflow.

1.2 Purpose

- The purpose of this **User Acceptance Testing (UAT)** document is to:
- **I.** Verify that the **Leave Approval System** functions correctly according to the defined requirements and user stories.
- **II.** Ensure that the **Leave Request** form captures all required employee information, including employee details, leave type, leave dates, and reason for leave.
- **III.** Validate that the **Flow Designer** workflow automatically routes leave requests to the appropriate reporting **Manager** for approval.
- **IV.** Confirm that managers can successfully **approve or reject** leave requests and that the leave status is updated correctly in the system.
- **V.** Test that **email notifications** are sent automatically at every stage of the leave process, including leave request submission, approval, and rejection.

- **VI.** Verify that the implemented **Business Rules** correctly validate leave dates, calculate leave duration, and prevent duplicate leave requests.
- **VII.** Confirm that all leave request records are accurately stored in the **Leave Request** table and can be viewed and managed within the ServiceNow platform.
- **VIII.** Ensure that the overall solution achieves the business objective of reducing manual paperwork, improving the efficiency of the leave approval process, increasing transparency, and providing accurate leave management within the organization.

2. IDEATION PHASE

2.1 Problem Statement

Date	22 June 2026
Team ID	
Project Name	Leave Approval System
Maximum Marks	2 Marks



Final Problem Statement

I am an employee trying to apply for leave and receive approval quickly, but the existing leave management process is manual, time-consuming, and difficult to track because there is no centralized system or automated approval workflow. This causes delays in leave approval, poor communication between employees and managers, and a lack of transparency regarding the status of leave requests.

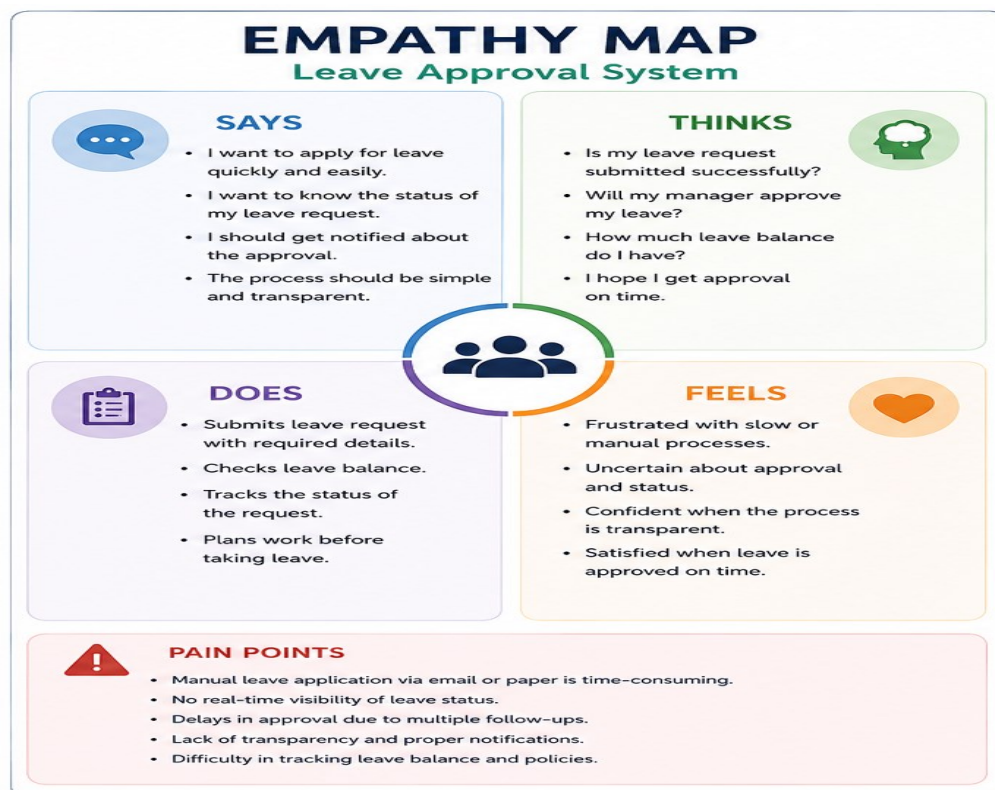
Solution

The **Leave Approval System** developed on the **ServiceNow** platform addresses these challenges by providing a centralized leave request application where employees can submit leave requests online. The system uses **Flow Designer** to automatically route requests to the reporting manager for approval. Based on the manager's decision, the leave request is approved

or rejected, the leave status is updated automatically, and email notifications are sent to the employee. Additionally, **Business Rules** validate leave dates, calculate leave duration, and prevent duplicate leave requests, making the leave management process faster, more accurate, and more transparent.

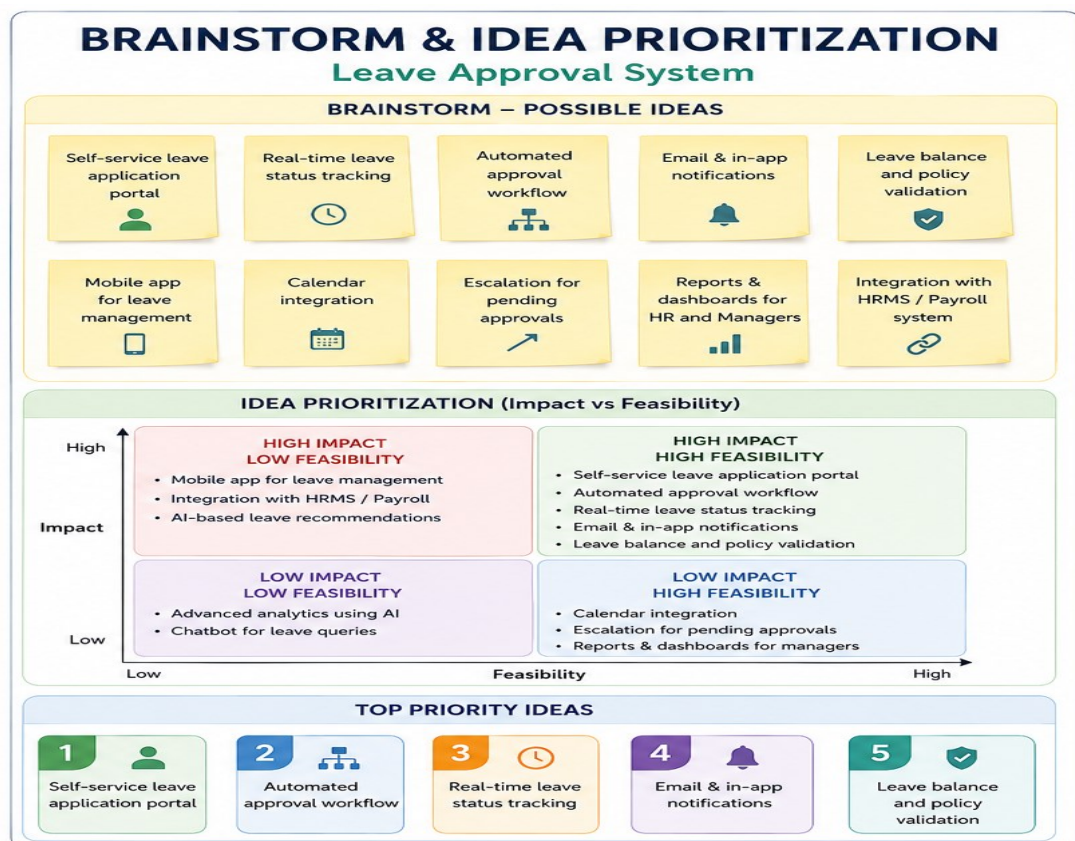
2.2 Empathy Map canvas

Date	22 June 2026
Team ID	
Project Name	Leave Approval System
Maximum Marks	4 Marks



2.3 Brainstorm & Idea Prioritization

Date	22 June 2026
Team ID	
Project Name	Leave Approval System
Maximum Marks	4 Marks



3. REQUIREMENT ANALYSIS

3.1 Solution Requirements (Functional & Non-functional)

Date	22 June 2026
Team ID	
Project Name	Leave Approval System
Maximum Marks	4 Marks

Functional Requirements

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Leave Request Form	Create a Leave Request table and form with fields such as Employee, Employee ID, Leave Type, Start Date, End Date, Reason, Manager, and Leave Status.
FR-2	Leave Request Submission	Allow employees to submit leave requests. Automatically create and store the leave request record in the Leave Request table.
FR-3	Approval Workflow	Configure Flow Designer to automatically send leave requests to the reporting manager for approval. Update the leave status based on the manager's decision.
FR-4	Email Notifications	Automatically send email notifications when a leave request is submitted, approved, or rejected. Notify the manager and employee at appropriate stages.
FR-5	Business Rules	Validate leave dates, calculate leave duration automatically, and prevent duplicate leave requests before saving the record.
FR-6	Leave Status Tracking	Allow employees and managers to view the current leave request status (Pending, Approved, Rejected) and maintain complete leave history.

Non-Functional Requirements

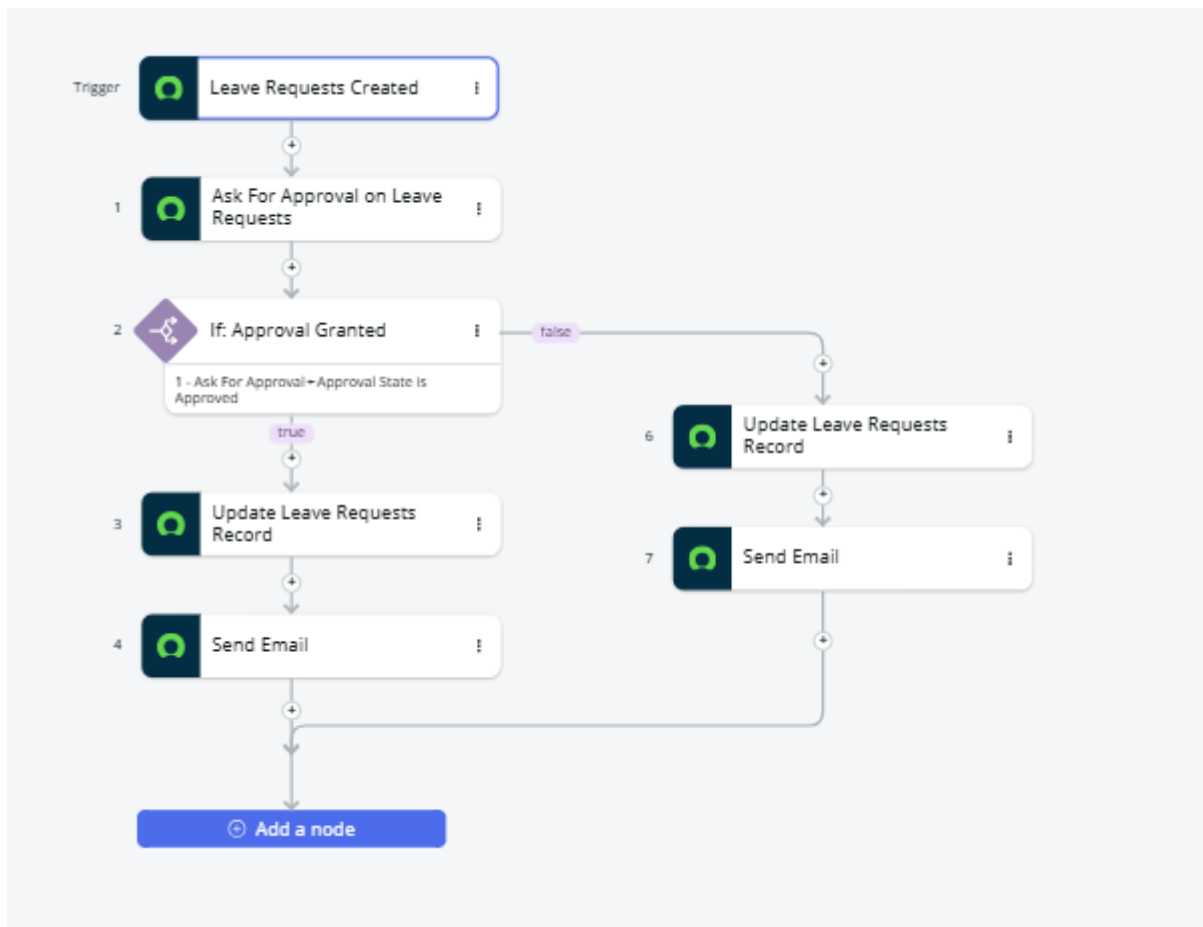
Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Leave Request form should have a simple, user-friendly interface that enables employees to apply for leave easily without technical knowledge.
NFR-2	Security	Only authorized employees can submit leave requests, and only assigned managers can approve or reject requests. Access is controlled using ServiceNow roles and permissions.
NFR-3	Reliability	The Flow Designer workflow should execute successfully for every leave request without manual intervention, ensuring consistent approval processing.
NFR-4	Performance	Leave requests, approvals, status updates, and email notifications should be processed within a few seconds after submission or manager action.
NFR-5	Availability	The ServiceNow Personal Developer Instance (PDI) should be available to employees and managers at any time to submit, review, and manage leave requests.
NFR-6	Scalability	The Leave Approval System should support additional leave types, multi-level approvals, more departments, and integration with HR systems without major redesign.

3.2 Data Flow Diagram & User Stories

Date	22 June 2026
Team ID	
Project Name	Leave Approval System
Maximum Marks	4 Marks

Data Flow Diagrams:



3.3 Technology Stack (Architecture & Stack)

Date	22 June 2026
Team ID	
Project Name	Leave Approval System
Maximum Marks	4 Marks

Technical Architecture:

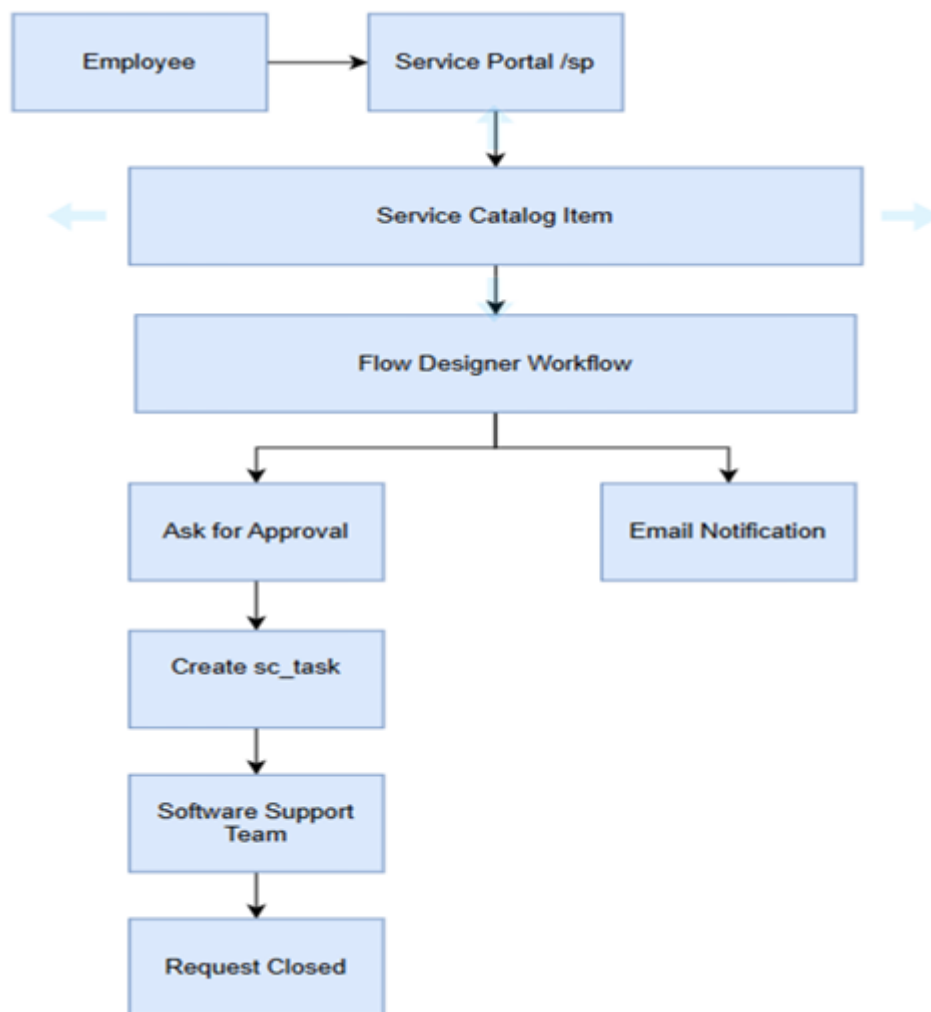


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employees submit leave requests through the Leave Request form.	ServiceNow Form Designer
2	Workflow Automation	Routes leave requests to the manager and updates leave status automatically.	ServiceNow Flow Designer
3	Business Rules	Validates leave dates, calculates leave duration, and prevents duplicate requests.	ServiceNow Business Rules (JavaScript / Glide API)
4	Database	Stores leave request details, employee information, manager approvals, and leave status.	ServiceNow Custom Table (Leave Request)

Table-2: Application Characteristics

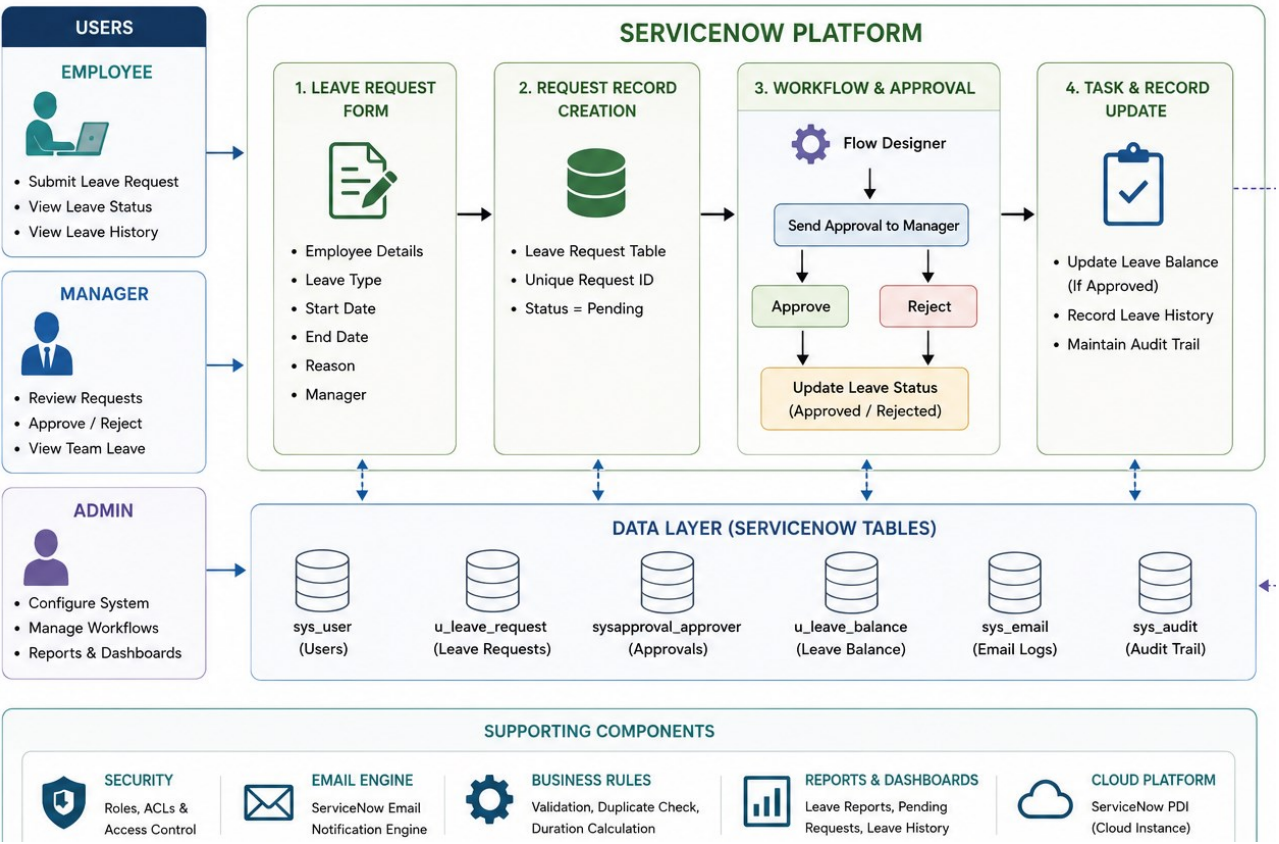
S.No	Characteristics	Description	Technology
1	Usability	Simple and user-friendly interface for submitting leave requests.	ServiceNow Form Designer
2	Security	Role-based access ensures only authorized users can submit or approve leave requests.	ServiceNow Roles & ACLs
3	Availability	Cloud-hosted ServiceNow instance available 24x7.	ServiceNow PDI
4	Performance	Leave requests, approvals, and notifications are processed quickly using automated workflows.	ServiceNow Flow Designer

4. PROJECT DESIGN

4.1 Problem – Solution Fit

Date	22 June 2026
Team ID	
Project Name	Leave Approval System
Maximum Marks	4 Marks

PROJECT DESIGN – LEAVE APPROVAL SYSTEM (ServiceNow)



4.2 Proposed Solution

Date	22 June 2026
Team ID	
Project Name	Leave Approval System
Maximum Marks	4 Marks

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Employees in an organization request network access (VPN, WiFi, LAN, Firewall, IP Address) through manual emails and phone calls, causing delays, lack of visibility, compliance issues with network security policies, and increased workload for the network team with no standardized process.
2.	Idea / Solution description	A ServiceNow-based Network Request Management System using a standardized Service Catalog item, Flow Designer workflow for automated approvals, auto-created fulfilment tasks for the Network Team, role-based access control for exception handling, and email notifications at every stage all accessible through the ServiceNow Service Portal.
3.	Novelty / Uniqueness	Unlike manual email-based processes, this solution provides a fully automated end-to-end request lifecycle within ServiceNow from submission to task fulfilment with zero manual coordination, real-time tracking, and built-in role-based access enforcement through Access Control Lists.
4.	Social Impact / Customer Satisfaction	Employees get faster network access with full visibility into request status. Network Engineers spend less time on manual coordination and more time on critical fulfilment tasks. Managers get structured, policy-compliant approval requests. Overall IT service satisfaction improves significantly.
5.	Business Model (Revenue Model)	This is an internal IT automation project for an organization. Cost savings are achieved by reducing network/IT support hours spent on manual request handling, minimizing unauthorized or untracked access changes, and ensuring policy compliance directly reducing operational costs.
6.	Scalability of the Solution	The solution is built on ServiceNow's native Flow Designer and Service Catalog making it easily scalable to handle additional request types, departments, and approval groups. Additional catalog items, workflow branches, and notification rules can be added without redesigning the existing system.

4.3 SOLUTION ARCHITECTURE

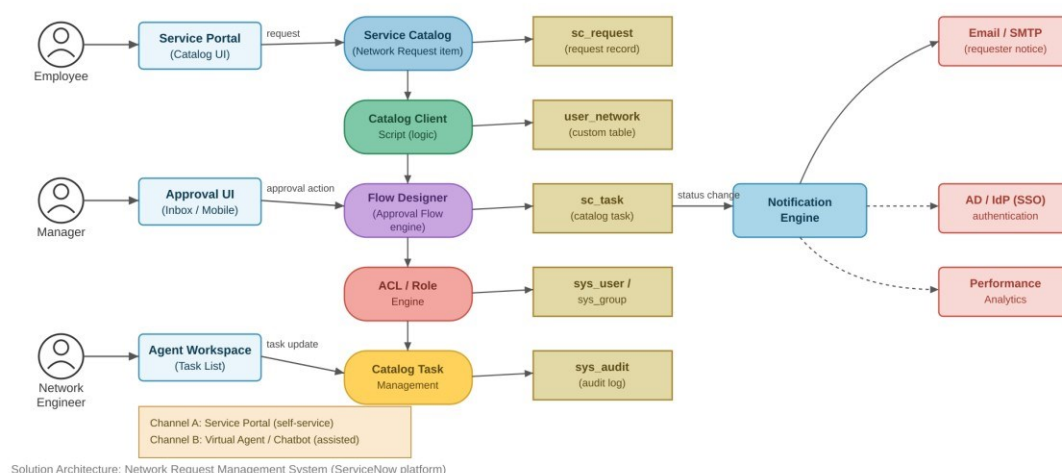
Date	22 June 2026
Team ID	
Project Name	Leave Approval System
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram -Network Request Management System:



5. PROJECT PLANNING AND SCHEDULING

5.1 PROJECT PLANNING

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint1	Network Request Submission	USN-1	As an employee, I can select “Network Request” from the Service Catalog and choose VPN, Wi-Fi, LAN, Firewall, or IP Address as my request type.	3	High
Sprint1		USN-2	As an employee, I can see only the fields relevant to my selected request type, since irrelevant fields are hidden automatically.	2	High
Sprint1		USN-3	As an employee, I can attach supporting documents to my request so the approver has the context needed to decide.	1	Medium
Sprint2	Approval Workflow	USN-4	As a manager, I can review a submitted network request with requester details and justification before deciding.	2	High
Sprint2		USN-5	As a manager, I can approve or reject a request with comments so the requester understands the decision.	3	High

Sprint2	Task Creation & Assignment	USN-6	As the system, I can autcreate a Catalog Task and assign it to the Network Team group once a request is approved.	3	High
Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority
Sprint2		USN-7	As a network engineer, I can view catalog tasks assigned to the Network Team so I know which requests need action.	1	Medium
Sprint3	Fulfilment & Status Tracking	USN-8	As a network engineer, I can update a task to Work in Progress so the requester and manager can track active work.	2	Medium
Sprint-3		USN-9	As a network engineer, I can mark a task as Closed Complete once access is granted so the request lifecycle finishes correctly.	2	High
Sprint3		USN-10	As an employee, I can view the real-time status of my submitted request (Open / WIP / Closed Complete).	2	High
Sprint3	Notifications	USN-11	As an employee, I can receive an email notification once my request is completed so I don't need to follow up manually.	1	High
Sprint1	User & Role Management	USN-12	As a system administrator, I can create users and assign roles (Employee, Manager, Network Engineer, Administrator) so access is controlled correctly.	3	High
Sprint1	Group Management	USN-13	As a system administrator, I can create and maintain the Network Team group so approved tasks route to the right people.	1	Medium

Sprint-2	Data Management	USN-14	As a system administrator, I can manage the user_network custom table so network request data is stored and retrievable when needed.	2	Medium
----------	-----------------	--------	--	---	--------

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	10	5 Days	20 June 2026	24 June 2026	10	24 June 2026
Sprint-2	12	5 Days	25 June 2026	29 June 2026	12	29 June 2026

Velocity:

Total Story Points = Sprint 1 + Sprint 2 = 10 + 12 = 22

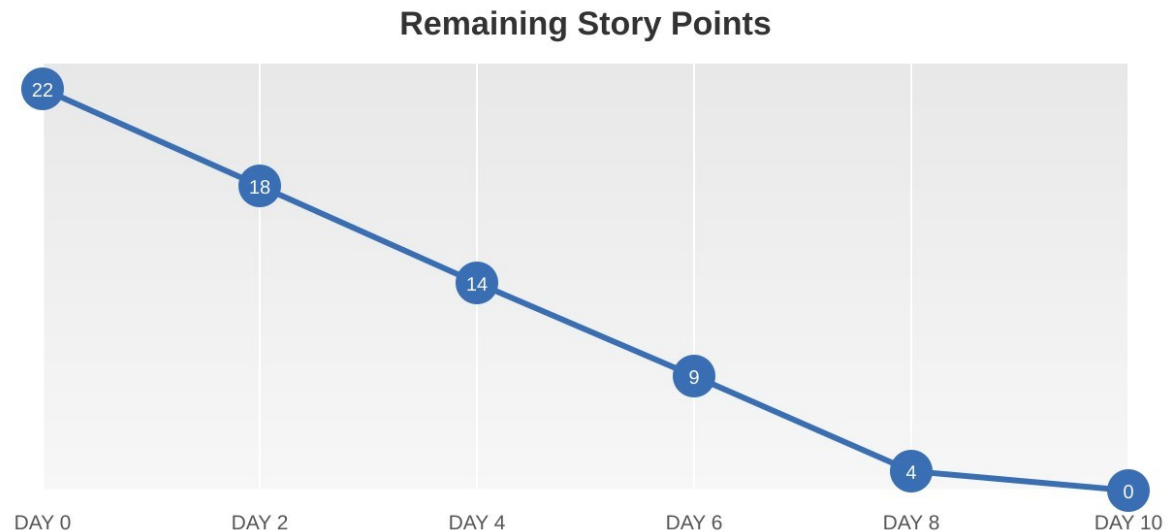
Number of Sprints = 2

Velocity = Total Story Points ÷ Number of Sprints = 22 ÷ 2 = 11 Story Points per Sprint

The team's velocity is 11 Story Points per Sprint.

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.



6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 USER ACCEPTANCE TESTING (UAT)

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	4 Marks

Project Overview:

Project Name: Leave Approval System

Project Description: A ServiceNow-based application that automates the submission, approval, fulfilment, and tracking of employee network access requests (VPN, Wi-Fi, LAN, Firewall, and IP Address access) through a Service Catalog item and an automated Flow Designer approval workflow.

Project Version: 1.0

Testing Period: 20 June 2026 to 28 June 2026

Testing Scope:

Features and functionalities to be tested:

- Network Request catalog item, including dynamic field display by Request Type
- Manager approval and rejection workflow (Network Request Approval Flow)
- Catalog Task auto-creation and assignment to the Network Team
- Task fulfilment and status updates (Work in Progress, Closed Complete)
- Email notifications to the requester on completion or rejection

User stories / requirements to be tested: USN-1 through USN-14 (see Solution Requirements document).

Testing Environment:

URL/Location: <https://dev391575.servicenow.com>

Credentials (if required): 927623bit116@mkce.ac.in

Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	Submit a VPN access request	1. Log in as Employee 2. Open Network Request catalog item 3. Select Request Type = VPN Access 4. Fill required fields and submit	Request is created with status Open and the correct fields for VPN Access are shown	As expected	Pass
TC-002	Dynamic field visibility on Request Type change	1. Open Network Request catalog item 2. Switch Request Type between VPN, Wi-Fi, LAN, Firewall, IP	Only fields relevant to the selected request type are shown; others are hidden automatically	As expected	Pass
TC-003	Attach supporting document to request	1. Open Network Request catalog item 2. Attach a file 3. Submit the request	Attachment is saved and visible on the submitted request record	As expected	Pass
TC-004	Manager approves a request	1. Log in as Manager 2. Open pending approval 3. Click Approve with a comment	Request status updates to Approved; a Catalog Task is auto-created and assigned to the Network Team	As expected	Pass

TC-005	Manager rejects a request	1. Log in as Manager 2. Open pending approval 3. Click Reject with a comment	Request status updates to Rejected; requester receives an email with the rejection comment	As expected	Pass
TC-006	Network Engineer fulfils a task	1. Log in as Network Engineer 2. Open assigned Catalog Task 3. Update state to Work in Progress 4. Close task as Complete	Task and parent request status update to Closed Complete; requester receives a completion email	As expected	Pass
TC-007	Unauthorized user attempts to approve a request	1. Log in as Employee (nonmanager) 2. Attempt to open an approval record directly via URL	Access is denied due to ACL restrictions	As expected	Pass

Bug Tracking:

Bug ID	Bug Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	Firewall Access fields remain visible after switching to VPN Access request type	1. Open Network Request catalog item 2. Select Request Type = Firewall Access 3. Switch Request Type to VPN Access	Medium	Open	Client script condition for Firewall-specific fields needs an additional onChange check
BG-002	Approval email not sent to Manager when request is submitted after business hours	1. Submit a request after 6 PM 2. Check Manager's inbox	High	In Progress	Notification trigger appears to depend on a scheduled job; recommend switching to eventdriven trigger
BG-003	Catalog Task not auto-assigned to Network Team for IP Address Request type	1. Submit and approve an IP Address Request 2. Check Catalog Task assignment group	High	Closed	Fixed by correcting the assignment group reference in Flow Designer for this request type

Sign-off:

Tester Name: SUBHA R

Date: 28 June 2026

Signature: SUBHA R

Notes:

- Ensure that all test cases cover both positive and negative scenarios (e.g. valid approvals and unauthorized access attempts).
- Encourage testers to provide detailed feedback, including any suggestions for improvement.
- Bug tracking should include details such as severity, status, and steps to reproduce.
- Obtain sign-off from both the project manager and product owner (or equivalent Network Team lead) before proceeding with deployment.

6.2 ACCEPTANCE TESTING UAT Execution & Report Submission

Date	22 June 2026
Team ID	
Project Name	Network Request Management System
Maximum Marks	4 Marks

1. Purpose of Document

The purpose of this document is to briefly explain the test coverage and open issues of the Network Request Management System project at the time of release to User Acceptance Testing (UAT).

2. Defect Analysis

This report shows the number of resolved or closed bugs at each severity level, and how they were resolved.

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	4	2	1	1	8
Duplicate	1	0	1	0	2
External	0	1	0	1	2
Fixed	6	3	2	9	20

Not Reproduced	0	0	1	0	1
Skipped	0	0	1	1	2
Won't Fix	0	2	1	0	3
Totals	11	8	7	12	38

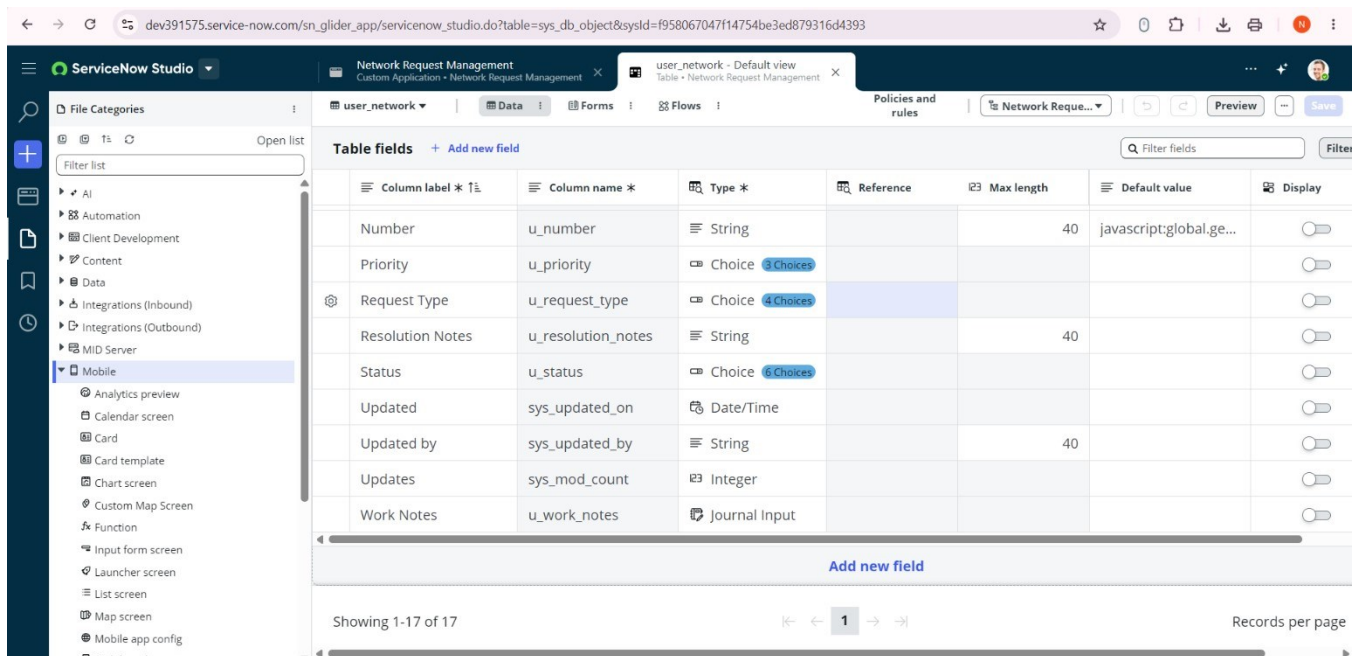
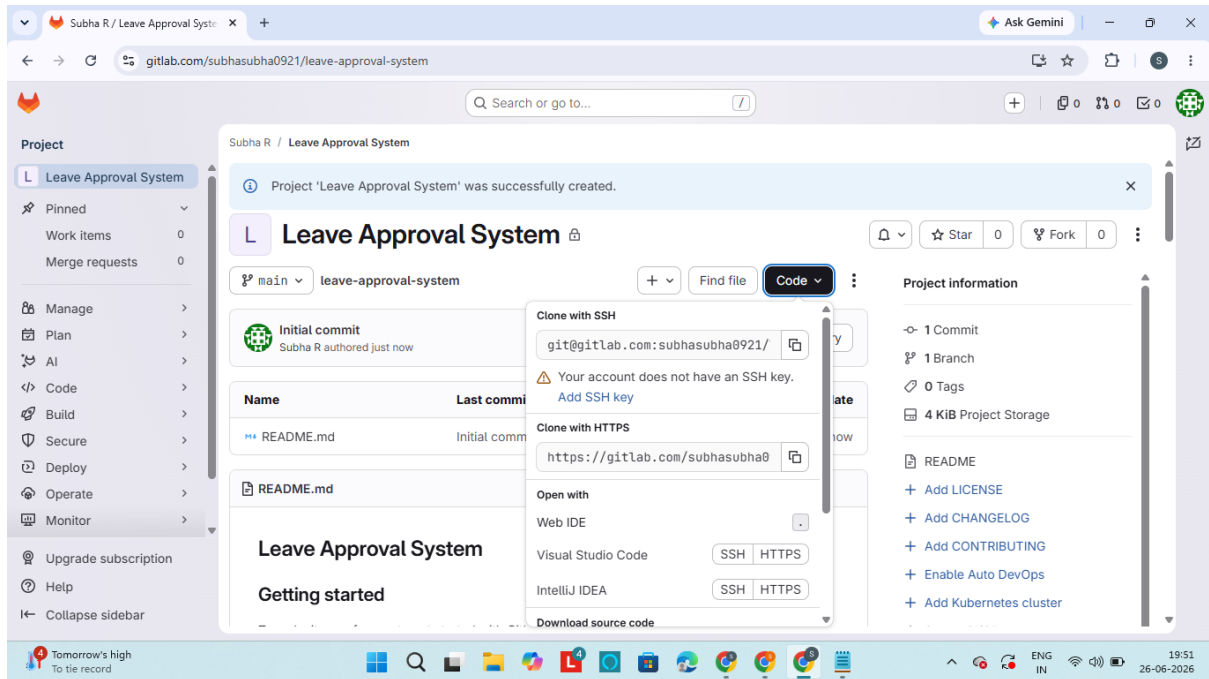
3. Test Case Analysis

Section	Total Cases	Not Tested	Fail	Pass
Network Request Catalog Form	14	0	1	13
Approval Workflow (Flow Designer)	10	0	0	10
Catalog Task Management	9	0	1	8
Roles & Access Control (ACL)	6	0	0	6
Email Notifications	5	0	1	4
user_network Custom Table	4	1	0	3
Reporting & Performance Analytics	3	0	0	3
Totals	51	1	3	47

This report shows the number of test cases that have passed, failed, and untested.

7. RESULTS

7.1 OUTPUT SCREENSHOTS



dev391575.service-now.com/now/nav/ui/classic/params/target/sys_user_group_list.do%3Fsysparm_query%3DnameSTARTSWITHnetwork%26sysparm_first_row%3D1%26sys...

servicenow

AllFavoritesHistoryWorkspacesAdmin

Groups

Search

Search

Actions on selected rows...New

All > Name starts with network

Name	Description	Active	Manager	Parent	Updated
network					
Network		true	Bow Ruggeri	(empty)	2023-06-06 19:02:08
Network Admin	Full control of the application	true	(empty)	(empty)	2026-06-28 00:20:58
Network Approvers	Approve network requests	true	(empty)	(empty)	2026-06-28 00:19:36
Network CAB Managers	Group to represent the CAB Managers for ...	true	(empty)	(empty)	2026-03-31 01:36:33
Network Requesters	Employees submit requests	true	(empty)	(empty)	2026-06-28 00:20:27
Network Team	Engineers work on network tasks	true	(empty)	(empty)	2026-06-28 00:18:50

1 to 6 of 6

dev391575.service-now.com/now/nav/ui/classic/params/target/sys_user_role_list.do%3Fsysparm_query%3DnameSTARTSWITHnetwork%26sysparm_first_row%3D1%26sysp...

servicenow

AllFavoritesHistoryWorkspacesAdmin

Roles

Search

Search

Actions on selected rows...New

All > Name starts with network

Name	Description	Elevated privilege
network		
network_admin	Full access to the application	false
network_approver	Manager can approve/reject requests	false
network_engineer	Network team can work on tasks	false
network_requester	Employee can submit network requests	false
network_user		false

1 to 5 of 5

dev391575.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do%3Fsysparm_query%3Duser_nameSTARTSWITHEmployee1%26sysparm_first_row%3D1...

servicenowAllFavoritesHistoryWorkspacesAdminUsers

Search

UsersNameSearch

Actions on selected rows...New

All > User ID starts with employee1

User ID	Name	Email	Active	Created	Updated
employee1	Employee One	balu40709@gmail.com	true	2026-06-28 00:37:56	2026-06-28 00:37:56

1 to 1 of 1

servicenowAllFavoritesHistoryWorkspacesAdminUsers

Search

UsersNameSearch

Actions on selected rows...New

All > User ID starts with manager1

User ID	Name	Email	Active	Created	Updated
manager1	Manager One	balu40709@gmail.com	true	2026-06-28 00:43:56	2026-06-28 00:43:56

1 to 1 of 1

dev391575.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do%3Fsysparm_query%3Duser_nameSTARTSWITHEngineer1%26sysparm_first_row%3D1%26sysparm_max%3D10

servicenow

AllFavoritesHistoryWorkspacesAdmin

Users

Search

Actions on selected rows...New

All > User ID starts with Engineer1

User ID	Name	Email	Active	Created	Updated
Engineer1	Search	Search	Search	Search	Search
engineer1	Engineer One	balu40709@gmail.com	true	2026-06-28 00:45:37	2026-06-28 00:45:37

1 to 1 of 1

dev391575.service-now.com/now/nav/ui/classic/params/target/sys_user_list.do%3Fsysparm_query%3Duser_nameSTARTSWITHAdmin%26sysparm_first_row%3D1%26sysparm_max%3D10

servicenow

AllFavoritesHistoryWorkspacesAdmin

Users

Search

Actions on selected rows...New

All > User ID starts with admin1

User ID	Name	Email	Active	Created	Updated
admin1	Admin One	balu40709@gmail.com	true	2026-06-28 00:46:45	2026-06-28 00:46:45

Catalog Items

Name

Search

Actions on selected rows...

New

All > Type != Bundle > Class != Order guide > Type != Package > Class != Content Item > Published item is empty > Name starts with Network Request

Q

Name ▲

Short description

Active

Roles

Catalogs

Category

Price

Type

Updated

Network Request

Search

Search

Search

Search

Search

Search

!=Package

Search

Network Request

Request for network access or configuration

true

Service Catalog

Network Standard Changes

\$0.00

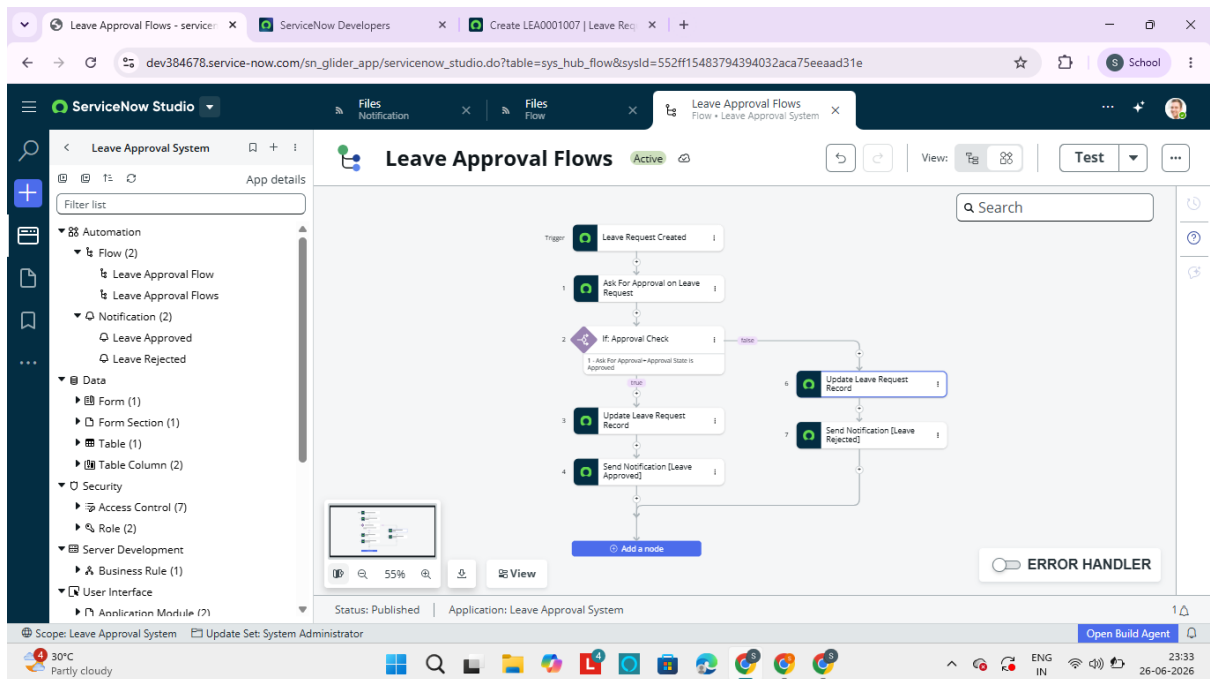
Item

2026-06-28 02:08:17

Activate

Deactivate

1 to 1 of 1



dev391575.service-now.com/sp?sysparm_stack=no

servicenow Knowledge Catalog Requests System Status Cart Tours System Administrator

How can we help?

How can we help?

Request Something
Browse the catalog for services and items you need

Knowledge Base
Browse and search for articles, rate or submit feedback

Get Help
Contact support to make a request, or report a problem

Current Status
No system is reporting an issue
More information...

My Assessments and Surveys
No assessments or surveys for you at the moment

My Approvals
You have no pending approvals

Announcements
Employee Center is available to you

My Open Incidents
Email server is down.
INC0009005 • about an hour ago

Top Rated Articles

Leave Approval Flows - servicenow ServiceNow Developers Create LEA0001007 | Leave Request

dev384678.service-now.com/now/nav/ui/classic/params/target/x_1974159_leave_0_leave_request.do%3Fsys_id%3D-1

servicenow All Favorites History Leave Request - Create LEA0001007 Search School

Leave Request
New record

End Date cannot be earlier than Start Date

Invalid insert

Employee Karthick

Employee ID 092634

Leave Type Sick Leave

Start Date 2026-06-30

End Date 2026-06-28

Reason Sick leave reason

status Approved

Submit

30°C Partly cloudy 23:29 26-06-2026

dev391575.service-now.com/sp?id=sc_request&is_new_order&table=sc_request&sys_id=6f221b74473d4754be3ed879316d43c4

System Administrator

firewall.png Attached

Home > Request Summary - REQ0010001

Search Catalog

Submitted : 2026-06-28 05:15:56
Request Number : REQ0010001
Requested for : Admin One
Estimated Delivery : 2026-06-30

Item	Delivery Date	Stage	Price (each)	Quantity	Total
Network Request	2026-06-30	Request Approval	---	2	---

Total: \$0.00

Approval - Student: STU0001019

Search

Update Delete

Approval Student: STU0001019

Approver System Administrator

Approving Student: STU0001019

State Rejected

Approval Reason

Comments

Post

Activities: 4

System Administrator

no need

Comments • 2026-06-28 05:29:24

System Administrator

State Rejected was Approved

Field changes • 2026-06-28 05:29:24

System Administrator

State Approved was Requested

Field changes • 2026-06-01 21:33:29

System Administrator

Approver System Administrator
State Requested

Field changes • 2026-06-01 21:50:34

Update Delete

Created	Recipients	Subject	State
2026-06-28 01:00:15	aileen.mottern@example.com	Restocking Request For Dell Inc. PowerEdge M710HD Blade Server	Ready
2026-06-28 05:16:13	admin@example.com, balu40709@gmail.com	Request REQ0010001 was approved	Ready
2026-06-28 01:00:15	aileen.mottern@example.com	Restocking Request For Fujitsu 1TB Hybrid Solid State Drive	Ready
2026-06-28 01:00:13	aileen.mottern@example.com	Restocking Request For APC 42U 3100 SP2 NetShelter Rack	Ready
2026-06-28 00:00:13	admin@example.com	Daily job to fetch Email Indicator Data and Email Notifications created completed with error	Ready
2026-06-28 05:16:03	fred.juddy@example.com	Catalog Task SCTASK0010002 has been assigned to group Software	Ready
2026-06-28 05:16:03	admin@example.com, balu40709@gmail.com	Request REQ0010001 was created	Ready

8. ADVANTAGES AND DISADVANTAGES

Advantages

The Network Request Management System offers several advantages by automating the complete network request process. It reduces manual paperwork and minimizes the use of emails for requesting network services. The system provides a centralized platform where employees can easily submit requests through the Service Catalog. Automated approval workflows reduce processing time and improve operational efficiency. Tasks are automatically assigned to the appropriate network team, eliminating manual task allocation. Users can track the status of their requests in real time, which improves transparency and communication. Role-based access control enhances system security by allowing only authorized users to perform specific actions. Email notifications keep requesters informed throughout the request lifecycle, ensuring better user satisfaction. The system also maintains a complete history of requests, making auditing and reporting easier.

Disadvantages

The system depends entirely on the ServiceNow platform and requires proper licensing and internet connectivity. Initial configuration and workflow development require technical knowledge of ServiceNow administration and development. If workflows or

approvals are configured incorrectly, requests may not be processed as expected. Organizations may also need to train employees before they can effectively use the system. Integration with external network management tools may require additional customization and development effort.

9. CONCLUSION

The Network Request Management System successfully automates the complete lifecycle of network service requests using the ServiceNow platform. Employees can easily submit network requests through the Service Catalog, while managers approve or reject requests through an automated approval process. Once approved, the system automatically creates tasks for the Network Team, updates request statuses, and sends email notifications to users. The project improves efficiency, reduces manual effort, enhances transparency, and provides better tracking of network requests. Overall, the system demonstrates the practical implementation of ServiceNow Administration, Service Catalog, Flow Designer, Client Scripts, Workflow Automation, and Notification Management.

10. FUTURE SCOPE

The project can be further enhanced by implementing multiple approval levels based on request type or business criticality. Service Level Agreements (SLAs) can be added to monitor request completion time and improve service quality. The application can be integrated with Active Directory for automatic user account provisioning and network access management. Additional communication methods such as Microsoft Teams, Slack, or SMS notifications can be integrated. AI-powered recommendations can assist users in selecting appropriate request types and predicting approval outcomes. Advanced dashboards and analytics can be developed to monitor request trends, team performance, and SLA compliance. Mobile support can also be enhanced to allow users to submit and track requests through the ServiceNow mobile application.

11. APPENDIX

Modules Developed:

- User Management
- Role Management
- Group Management
- Service Catalog
- Catalog Variables
- Catalog Client Script
- Flow Designer
- Approval Workflow
- Catalog Task Creation
- Email Notifications

Tools Used:

- ServiceNow Developer Instance
- Flow Designer
- Service Catalog
- JavaScript (Catalog Client Script)

Source Code:

```
function onChange(control, oldValue, newValue, isLoading) {  
  
    if (isLoading || newValue ==  
    "") {        return;
```

```
}  
  
    if (newValue == 'vpn') {  
g_form.setDisplay('access_level', true);  
    } else {  
        g_form.setDisplay('access_level', false);  
    }  
}
```

Github Link

<https://github.com/Subha23456/Leave-Approval-System>

Demo Link

<https://drive.google.com/drive/u/0/folders/1Ea1SPYaqeIOD0zW9bRtfKtL6eqIVBoTd>